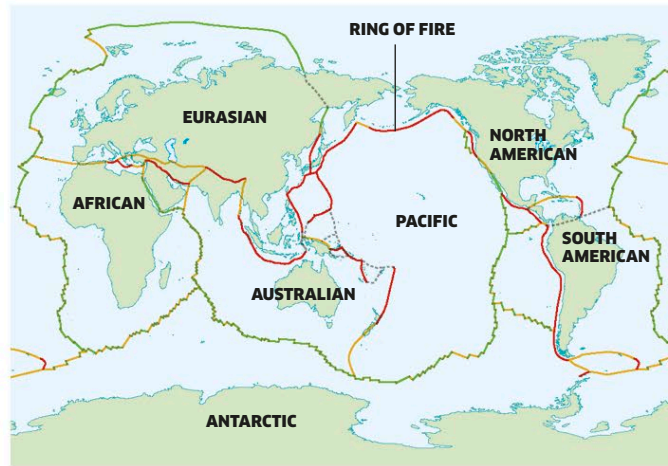


## Tectonic plates

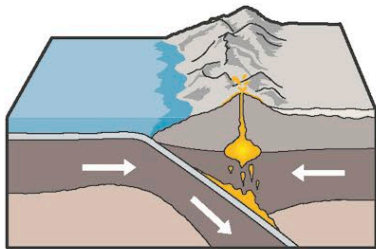
There are seven large plates and numerous medium-sized and smaller plates, which roughly coincide with the continents and oceans. The Ring of Fire is a zone of earthquakes and volcanoes around the Pacific plate from California in the northeast to Japan and New Zealand in the southwest.

- **CONVERGENT**
- **DIVERGENT**
- **TRANSFORM**
- - - - **UNCERTAIN**



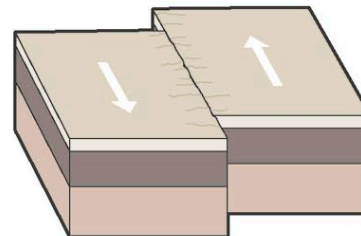
## Colliding continents

When continents collide, layers of rock are pushed up into mountain ranges. Continental convergence between the Indian subcontinent and Eurasian landmass formed the Himalayas.



### Convergent boundary

As two plates move toward each other, one plate moves down, or subducts, under the other and is destroyed. A deep-sea trench or chain of volcanoes may form, and earthquakes often occur.



### Transform boundary

When plate edges scrape past each other, earthquakes are frequent. The San Andreas fault in California is a famous example.

### Sliding plates

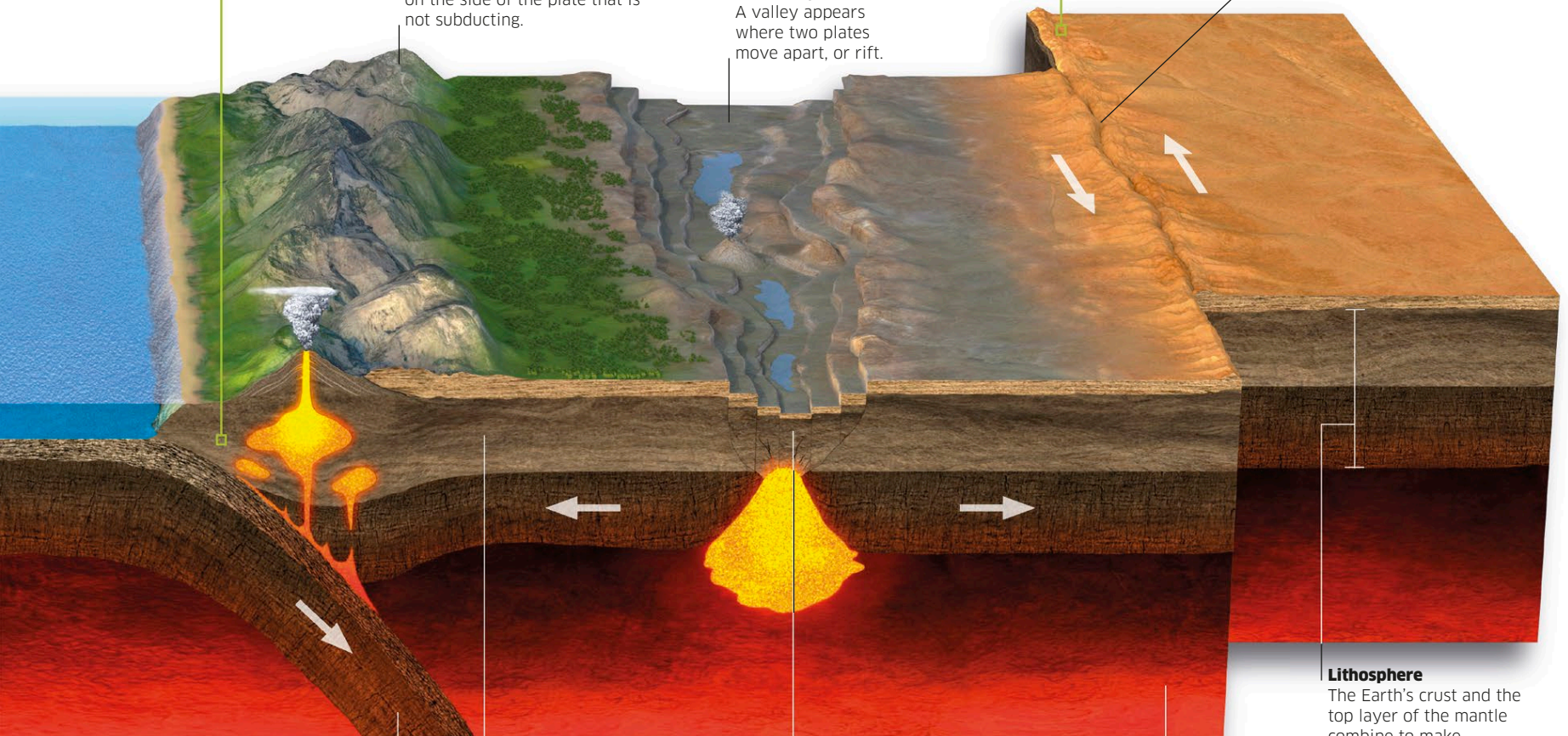
Plates sliding past each other may make earthquakes happen.

### Volcanic ranges

A chain of volcanoes develops on the side of the plate that is not subducting.

### Rift valley

A valley appears where two plates move apart, or rift.



### Oceanic-continental subduction

A thinner oceanic plate slides under the thicker continental plate at this boundary.

### Continental crust

The Earth's crust is thicker and less dense on land than under the oceans.

### Continental rift

When two continental plates move apart, they create a rift—as in East Africa's Rift Valley. Magma rises up through the gap, leading to volcanic activity.

### Asthenosphere

Temperature and pressure combine to make the rock in this layer semi-molten.

### Lithosphere

The Earth's crust and the top layer of the mantle combine to make the rigid lithosphere.